

Ms. Ref. No.: **KNOSYS-D-26-21490**

Title: *Effective Training Pressure Gates Recursive Knowledge Degradation in LLMs: A Multi-Axis Dose-Response Study*

Journal: *Knowledge-Based Systems* (Elsevier)

To: Dr. Hang Yu, Senior Editor

Re: Response to Reviewers — Revision Round 1

Date: 19 October 2026

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Dear Editor and Reviewers,

We thank you for your careful and constructive evaluation of our manuscript. Your comments prompted substantive revisions to the experimental account, statistical reporting, notation, and scope of our conclusions. We respond to each comment below and indicate the corresponding changes in the manuscript.

The most consequential correction concerns the original synthetic-exposure comparison. During revision, we identified that its baseline and intervention conditions had been evaluated through different pipeline paths, including differences in the construction of the baseline-correct question set ( $K_0$ ). Consequently, we no longer rely on that comparison as evidence for the previously reported intervention effect. The revised manuscript instead presents paired dose-response experiments conducted under consistent pipelines. On Qwen, reducing synthetic exposure attenuated the Generation 5–10 retention decline at the 50% reduction level. On Gemma 3, the corresponding intervention improved Generation 10 retention but did not attenuate the decline over that interval. We have revised the interpretation to preserve this distinction rather than proposing a common mechanism that the experiments do not establish.

We have also clarified the status of effective training pressure (ETP). It is now presented as an operational framework for organizing experimental conditions and their observed diagnostics, not as a validated universal scalar predictor or an architecture-independent threshold. Additional revisions distinguish controlled comparisons from descriptive contrasts, report the available seed-level uncertainty, define the notation used in the equations and figures, and state the limits of the factual-retention evaluation.

The responses that follow identify the action taken for each concern and direct the reader to the relevant part of the revised manuscript. Revised manuscript passages quoted in this letter appear in [blue](#).

Respectfully submitted on behalf of all authors,

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Corresponding Author

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## Conventions used in this response

- Reviewer comments are reproduced *in green italics*, verbatim, including the table and section numbers of the *original* submission.

- Text newly added to the manuscript appears “in blue”. Where a passage from the revised manuscript is reproduced in this letter, it is paraphrased or summarised; the relevant section is cited so that the exact wording can be located in the revision.
- Locations are given as section and page of the revised manuscript. Where the revision renumbered a table or section, both numbers are given at first mention, e.g. Table 7 of the original submission (now Table 9).
- All numerical values quoted here are generated from the archived per-seed results by `scripts/analysis/make_a` and are reproduced verbatim from its output.

## Experiment identifiers used in this response

The following identifiers appear in the revised manuscript and in this response. Table 3 of the revised manuscript maps the control experiments to the sources of variation they address.

| ID    | Experiment                                        | Backbone      | Seeds                      | Role                                          |
|-------|---------------------------------------------------|---------------|----------------------------|-----------------------------------------------|
| G1    | QLoRA rank sweep, NF4                             | Qwen 2.5 1.5B | 3 at $r=256$               | Primary dose-response                         |
| G2    | Pre-registered exposure dose-response, $r=10$     | Gemma 3 1B IT | 3 per dose                 | Cross-backbone replication                    |
| G3    | Precision control: bf16                           | Qwen 2.5 1.5B | 3 at $r=16$ ; 3 at $r=256$ | Rules out quantization artefact               |
| G4    | Data-order control: three new shuffle seeds       | Qwen 2.5 1.5B | 3 at $r=256$               | Rules out ordering artefact                   |
| G5    | Pre-registered prospective rank $\times$ LR probe | Qwen 2.5 1.5B | 1 per cell                 | Exploratory ordering test                     |
| B7    | Pre-registered exposure dose-response, $r=256$    | Qwen 2.5 1.5B | 5 per dose                 | Primary exposure evidence                     |
| C1–C5 | Exploratory intervention pilot (archived)         | Qwen 2.5 1.5B | 1                          | Superseded by B7; excluded from all estimates |

## Summary of changes

Forty review points were raised: three by the Editor, five by Reviewer 1, and thirty-two by Reviewer 2. Each is answered individually below. Points marked PART are addressed in part, with the residual limitation stated explicitly in Section 6. The table maps every point to the revision made and its location.

| Point         | Issue raised      | Revision made                                                                     | Location           |
|---------------|-------------------|-----------------------------------------------------------------------------------|--------------------|
| <b>Editor</b> |                   |                                                                                   |                    |
| ED-1          | Language editing  | Complete language revision; terminology standardized throughout                   | Throughout         |
| ED-2          | Submission format | Seven documents in Word; figures at 300 dpi (non-PDF); <code>.bib</code> included | Submission package |

| Point             | Issue raised                                                                               | Revision made                                                                                               | Location                  |
|-------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|---------------------------|
| ED-3              | List of changes                                                                            | This table plus an individual response to each point                                                        | This document             |
| <b>Reviewer 1</b> |                                                                                            |                                                                                                             |                           |
| R1-0              | Not a transferable predictor; single dataset; single-seed analyses; one exposure condition | Framework reframed as operational; scope and seed policy stated per claim                                   | §3.5, §3.8, §6            |
| R1-1              | Prospective calibration PART                                                               | Five-step calibration protocol specified; held-out validation declared not performed                        | §3.5, §6                  |
| R1-2              | Confound separation                                                                        | Controls G3 (precision) and G4 (data order); fresh optimizer per generation; design/observed split          | §3.9, Tab. 3, §4.6        |
| R1-3              | External validity PART                                                                     | Strict exact-match re-scoring (Tab. S2); scope restriction stated                                           | §3.2, §6, Tab. S2         |
| R1-4              | Exposure: Qwen versus Gemma 3                                                              | Pipeline inconsistency disclosed; two pre-registered dose-response experiments replace the pilot            | §3.7, §4.5                |
| R1-5              | Reproducibility                                                                            | Checkpoint revisions, prompts, seeds, masks, item-level vectors, raw generations and adapters deposited     | §3.3, App. A              |
| <b>Reviewer 2</b> |                                                                                            |                                                                                                             |                           |
| R2-0              | Framework conceptual; Gemma 4 uncited; limited scale                                       | Contribution restated as design and evidence; Gemma 4 cited and disambiguated; scale stated as a limitation | §1, §3.3, §6              |
| R2-1              | Abbreviations and notation (10 points)                                                     | All acronyms expanded; notation table rebuilt; set/cardinality and singular-value notation corrected        | Tab. 1, §3.2–3.6          |
| R2-2              | Malformed equations (4 points)                                                             | Glyph fault traced and corrected; Eq. 1–6 rewritten; glyph audit script included                            | Eq. 1–6, Tab. 9           |
| R2-3              | Internal inconsistencies (7 points)                                                        | Each corrected; threshold language unified; seed claims aligned                                             | Abstract, §4, §5          |
| R2-4              | Methodology (6 points)                                                                     | Inferential statistics added; per-backbone calibration proposed; “sharp” withdrawn; Gemma 4 sourced         | §3.5, §4, §6              |
| R2-5              | References (5 points)                                                                      | Duplicate arXiv resolved; year corrected; Gemma 4 added; preprint status declared; list harmonized          | <code>cas-refs.bib</code> |

## Editor

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### Editor, Submission requirements — Format of the revised submission:

*While submitting revision, please ensure to submit a list of changes or a rebuttal against each point that was being raised. The revised cover letter, response to reviewers, highlights, revised manuscript, credit author statement, author agreement and declaration of interest should all be in word format. Source file alone can be in word or Latex file and only one final version of manuscript is only required. The tables, figures and equations should be in editable format in the revised manuscript. Figures should be in good quality 300 dpi and not in pdf format. For LaTeX version, kindly ensure to check if supporting .bib files are also included.*

### Author response:

We sincerely thank the Editor for the detailed submission requirements and for the opportunity to revise and resubmit this manuscript to *Knowledge-Based Systems*. We have carefully addressed each of the seven requirements and confirm that all necessary documents have been prepared according to the journal's specifications.

Each requirement is addressed individually. Requirement (a) calls for a point-by-point rebuttal: both a Summary of Changes table and individual responses are provided. Requirement (b): all seven documents produced in `.docx`. Requirement (c): a single `LATEX` source with compiled PDF. Requirement (d): all tables are native `tabular`; equations are `amsmath`; no raster images. Requirement (e): figures re-exported at 300 dpi as PNG. Requirement (f): `cas-refs.bib` audited and included.

Each requirement is addressed as listed below.

| Requirement                                        | Compliance                                                                                                                                                                                 |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) List of changes or rebuttal against each point | Both: a summary-of-changes table maps all forty review points, followed by an individual response to each.                                                                                 |
| (b) Seven documents in Word                        | Cover letter, response to reviewers, highlights, revised manuscript, CRediT author statement, author agreement and declaration of competing interest are submitted in <code>.docx</code> . |
| (c) Source file; one final version                 | A single final <code>L<sup>A</sup>T<sub>E</sub>X</code> source is submitted with its compiled PDF.                                                                                         |
| (d) Editable tables, figures, equations            | Tables are native <code>tabular</code> environments; equations are <code>amsmath</code> environments; no table or equation is an image.                                                    |
| (e) Figures at 300 dpi, not PDF                    | All figures are submitted at 300 dpi as PNG/TIFF. No figure is submitted in PDF.                                                                                                           |
| (f) Supporting <code>.bib</code>                   | <code>cas-refs.bib</code> is included in the <code>L<sup>A</sup>T<sub>E</sub>X</code> submission package.                                                                                  |

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### Author action:

1. **Seven Word documents produced.** All seven files (cover letter, response, highlights, manuscript, CRediT, agreement, declaration) produced in `.docx`.

2. **Figures regenerated at 300 dpi in PNG.** All figures re-exported at 300 dpi; PDF pipeline replaced.
3. **Bibliography audited.** Duplicate arXiv ID resolved; two missing IDs added; Ref. [31] attribution corrected.
4. **Language revision.** American English throughout; terminology standardized (regime labels, ETP abbreviation introduced at first use in Section 3.5).

*Revise (including language editing). Provide point-by-point response to each reviewer comment in the cover letter.*

#### **Author response:**

Thank you for this guidance. We have edited the manuscript throughout for language and clarity, with particular attention to the description of experimental conditions, the distinction between results and interpretations, and the consistency of terminology across sections. We have also moderated claims that extended beyond the tested architectures or evaluation setting.

#### **Author action:**

The manuscript underwent a complete language revision prior to resubmission, covering grammar, terminology consistency, and the removal of residual non-English constructions. Terminology was standardized throughout (Homeostatic / Bounded / Degradative; the term “effective training pressure” is written in full at first use and wherever a claim is made about it; the abbreviation ETP is introduced explicitly in Section 3.5 and used thereafter only in the compound labels ETP-design and ETP-observed), and all abbreviations are now expanded at first use. Should the Editor consider a further round of professional language editing necessary, we will gladly commission Elsevier’s Author Services before acceptance. Every reviewer comment below is addressed by a dedicated revision, with the corresponding section and page cited.

#### **Reviewer 1**

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#### **Reviewer #1, General assessment — Scope of the evidence:**

*This paper studies recursive fine-tuning on synthetic data across Qwen 2.5 1.5B, Gemma 3 1B, and Gemma 4 E2B for ten generations. The systematic design, multi-seed headline experiments, and distributional diagnostics are valuable. However, the proposed pressure is not yet a transferable quantitative predictor, the main metric uses one short-answer dataset, several analyses are single-seed, and the exposure intervention is shown mainly for one Qwen condition. Major revision is required.*

#### **Author response:**

We sincerely thank Reviewer #1 for the thorough and balanced assessment of this work. The four reservations identified in the general evaluation directly shaped the priorities of the revision, and we are grateful for the level of engagement with the experimental design. Addressing these reservations has substantially improved the precision of the claims and the transparency of the limitations.

We accept all four reservations in full and have restructured the manuscript so that each is stated explicitly as a property of the evidence.

**Reservation 1: ETP is not yet a transferable quantitative predictor.** ETP is now described as an operational experimental framework, not a validated predictor. “The ETP index  $\Pi$  is an exploratory post-training summary; it is not available prospectively and has not been validated as a transferable quantitative predictor across architectures or datasets.” This sentence appears in Section 3.5, the Abstract, and §7.

**Reservation 2: The main metric uses one short-answer dataset.** Strict exact-match re-scoring of all saved predictions was added (supplementary Table S2). The evaluation scope is now bounded explicitly: “The evaluation is conducted on short-answer factual questions (TriviaQA); results for longer answers, multi-hop questions, or other domains constitute an open empirical question.”

**Reservation 3: Several analyses are single-seed.** The seed policy is now explicit at the level of individual claims (Section 3.7). Headline conditions use three to six independent seeds; intermediate ablations use one seed and are labeled descriptive; the two categories are never pooled.

**Reservation 4: The exposure intervention is shown for mainly one Qwen condition.** The single-seed C1/C5 pilot has been replaced by two pre-registered dose-response experiments (B7: Qwen, five seeds; G2: Gemma 3, three seeds), each at four dose levels within a single consistent evaluation pipeline.

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### Reviewer #1, Concern #1 — Prospective calibration of ETP:

*How can effective training pressure be quantitatively calibrated and prospectively tested on an unseen backbone or dataset?*

#### Author response:

We sincerely thank the reviewer for this critical and precisely formulated question. The question of prospective calibration is fundamental: it asks whether effective training pressure can be used as a predictive instrument before running experiments on a new backbone, or whether it is only a post-hoc explanatory label. We acknowledge that the original manuscript did not address this distinction with sufficient clarity, and we take full responsibility for having allowed the presentation of ETP to suggest a degree of predictive transferability that the experiments do not support. The revision corrects this at every level of the manuscript.

**What the original manuscript implied and why it is incorrect.** The original manuscript introduced the ETP index  $\Pi$  as a unified quantitative measure that combines adapter rank, learning rate, and effective rank into a single scalar. While this framing is useful for organizing the experimental observations post hoc, it implies that  $\Pi$  could be computed before training (using only the nominal design parameters) and used to predict where a new configuration will fall in the regime structure. This implication is incorrect. The effective rank component of  $\Pi$  is measured after training; it is not available prospectively. Furthermore, the attempted cross-architecture normalizations showed that the observed regime boundaries do not align across Qwen 2.5 1.5B and Gemma 3 1B: the boundary falls between effective ranks  $\sim 3$ –6 on Gemma 3 and between 50–88 on Qwen. A single scalar therefore cannot predict regime membership across backbones without a backbone-specific calibration step.

**What the revised manuscript says instead.** The revised manuscript distinguishes two classes of variable explicitly in Section 3.5. ETP-design variables are nominal adapter rank, learning rate, planned training steps, and synthetic data dose; these are fixed before training and directly manipulated in the factorial design. ETP-observed variables are effective rank and mean absolute weight drift  $d(t)$ ; these are measured after training as diagnostic proxies for how much pressure

was actually applied. The text now states: “The ETP index  $\Pi$  is an exploratory post-training summary of the variables manipulated in this study. It is not available prospectively and has not been validated as a transferable quantitative predictor across architectures or datasets.” This sentence appears in Section 3.5, is restated in the abstract, and is reflected in the conclusion.

**What a genuine prospective test would require.** A prospective calibration on an unseen backbone would require: (i) a pilot sweep over two nominal ranks spanning roughly one order of magnitude, conducted for five recursive generations with a single seed, recording effective rank per generation; (ii) a backbone-local calibration step estimating the regime boundary as the interval between the highest pilot configuration whose Gen 1–5 retention slope is indistinguishable from zero and the lowest whose slope is negative; (iii) pre-specification of two configurations on opposite sides of the calibrated interval, together with the outcome criterion and decision rule, before any further run; (iv) evaluation of both configurations to Gen 10 with at least three seeds; and (v) reporting of the pre-registered prediction, the observed classification, and any discrepancy, irrespective of direction. This protocol is added to Section 6 as a concrete target for future work, not as an experiment already completed.

**The G5 experiment and its scope.** The exploratory G5 experiment provides a narrower check: Gen 5 retention was ordered in the predicted direction across all three single-seed cells (96.2%, 89.7%, 85.9%), but one of the three pre-specified regime classifications was not met: the high-pressure cell was predicted Degradative and returned Bounded. No threshold was pre-specified for G5, and it did not evaluate an unseen backbone or dataset. The revised manuscript characterizes G5 as exploratory evidence for the ordering of regimes, not as a confirmatory prospective test.

#### Author action:

1. **ETP redefined (Section 3.5).** The distinction between ETP-design and ETP-observed variables was formalized. The following sentence was added to Section 3.5: “The ETP index  $\Pi$  is an exploratory post-training summary; it is not available prospectively and has not been validated as a transferable quantitative predictor across architectures or datasets.”
2. **Abstract and conclusion qualified (Abstract; Section 7).** The abstract now bounds the ETP claim: “We identify effective training pressure as an organizing principle for the observed regime structure; this framework is not yet a transferable quantitative predictor, as cross-architecture normalization did not align the observed boundaries.” The conclusion repeats this qualification.
3. **Prospective calibration protocol added (Section 6).** A five-step protocol specifying the requirements for a genuine prospective test on an unseen backbone was added to Section 6 as a direction for future work (see the five-step list above).
4. **G5 interpretation qualified (Section 4.6).** The G5 result is now described as “exploratory evidence for the ordering of regime boundaries, not a confirmatory prospective test: no threshold was pre-specified and the backbone was not held out from the design.” The classification discrepancy (high-pressure cell predicted Degradative, returned Bounded) is reported explicitly.

*Revised manuscript: Section 3.5 (ETP definition); Section 4.6 (G5 prospective test); Section 6 (prospective validation and limitations).*

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#### Reviewer #1, Concern #2 — Separating pressure effects from confounders:

*How are quantization, optimizer state, data order, and model-specific dynamics separated from the claimed pressure effects?*

## Author response:

We sincerely thank the reviewer for this methodologically important concern. Separating the contribution of individual pressure axes (adapter rank, learning rate, quantization, data composition) from one another and from model-specific dynamics is a genuine challenge in this factorial design, and the reviewer is correct that the original manuscript did not state these limitations with sufficient precision. The revision makes explicit what each control arm establishes, what remains confounded by construction, and what additional experiments would be required to disentangle the effects.

The revised manuscript identifies what each control can establish and what remains confounded. We address each confounder raised by the reviewer in turn, and for each we state explicitly what the available evidence supports and what it cannot establish.

**Quantization (NF4 versus bf16).** Group G3 compares Qwen at  $r = 16$  and  $r = 256$  under the same bf16 protocol with three paired seeds. The per-seed Generation 10 differences are +15, +13, and +15 questions out of  $|K_0| = 78$ , a mean difference of +18.4 percentage points. The exact paired sign-flip test gives  $p = 0.25$ . This directionally consistent, small-sample result supports the rank contrast under bf16 but does not establish a statistically conclusive or architecture-independent effect. The 17.7-percentage-point Qwen difference obtained by combining conditions from different protocols is descriptive only: rank, quantization, and aspects of the data-order protocol differ across those groups, and we do not use that comparison as a rank-only hypothesis test. The bf16 check is an implementation consistency check, not a complete causal account of the observed trajectories. The revised manuscript states explicitly: “The bf16 comparison (G3) establishes that the rank contrast is directionally consistent under quantization-matched conditions ( $p = 0.25$ ,  $N = 3$  pairs); it does not isolate the rank effect from quantization nor establish a statistically conclusive result.”

**Optimizer state.** Under the reported protocol, a fresh optimizer is instantiated for each recursive generation; optimizer state is therefore not carried across generations. We have made this explicit in Section 3.1 with the sentence: “A fresh adapter and optimizer are initialized at the start of every generation; no optimizer state is carried across generations.”

**Data order.** In Group G4, both the random seed and the data order changed relative to Group G1. The Generation 10 comparison ( $N = 3 + 3$ , permutation  $p = 0.70$ ) consequently does not isolate a data-order effect, and the non-significant result does not prove that order has no effect. We have added this caveat explicitly: “G4 changes both seed and data order simultaneously; the  $p = 0.70$  permutation result does not isolate either variable independently.”

**Model-specific dynamics.** For Gemma 3, the  $r = 4$  versus  $r = 16$  contrast involves five independent runs per condition. An independent-seed exact permutation test yields  $p = 0.0079$ , whereas a paired sign-flip analysis (using matched seeds where available) yields  $p = 0.0625$ . We report both results with their differing interpretations: the permutation test suggests the difference exceeds chance under independence, while the sign-flip result reflects the uncertainty from only three paired seeds. The revised manuscript states: “We report both  $p$ -values because the pairing status is not fully confirmed across conditions; the two tests bracket the inferential uncertainty without overclaiming in either direction.”

**Summary of what remains confounded.** The revision includes a dedicated paragraph in Section 4.2.1 that summarizes which contrasts isolate individual axes, which remain partially confounded, and what additional experiments (e.g., a design that varies rank alone while holding quantization, optimizer, data order, and seed fixed simultaneously) would be required to draw fully causal conclusions. We have added: “The available controls establish directional consistency across multiple configurations; they do not permit fully causal attribution of the observed effects to individual pressure axes in isolation.”

## Author action:

1. **Optimizer state documented (Section 3.1).** The sentence “A fresh adapter and optimizer are initialized at the start of every generation; no optimizer state is carried across generations” was added to Section 3.1.
2. **Quantization check scoped (Section 4.2.1).** The bf16 check is now described as an implementation consistency check, not a causal isolation. The  $p$ -value ( $p = 0.25$ ,  $N = 3$ ) and the limitation are stated explicitly.
3. **Data-order caveat added (Section 4.2.3).** The G4 comparison is described as simultaneously changing seed and data order; the  $p = 0.70$  result is qualified accordingly.
4. **Summary paragraph on confounding added (Section 4.2.1).** A paragraph listing the available controls, their limitations, and the experiments required for fully causal inference was added to Section 4.2.1.

*Revised manuscript: Sections 3.5 and 3.9 (variables and controls); Section 4.1 (rank sweep); Section 4.2.1 (Qwen comparisons); Section 4.3.1 (Gemma 3); Section 4.6 (G3 and G4).*

### Revised manuscript:

The revised manuscript implements these controls throughout Sections 3.5, 3.9, 4.1, 4.2.1, 4.3.1, and 4.6.

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### Reviewer #1, Concern #3 — External validity of the evaluation:

*Do the conclusions hold for longer answers, multi-hop questions, other domains, and larger factual evaluation sets?*

### Author response:

We sincerely thank the reviewer for raising the question of external validity. The concern is legitimate: the evaluation in the original manuscript was conducted on short-answer factual questions (TriviaQA) with exact-match scoring, and claims about longer answers, multi-hop questions, or other domains were not supported by the data. The revision explicitly scopes the findings to the evaluation protocol used and positions generalization as an open empirical question rather than an established result.

Our results do not establish generalization to those settings, and the revision says so explicitly. We address each dimension of the external validity concern separately.

**Scope of the evaluation dataset.** The evaluation uses a held-out set of 200 TriviaQA questions, disjoint from the 2,000-question synthetic training stream. Retention is measured as the fraction of questions in  $K_0$  (those answered correctly at generation 0) that remain correctly answered at generation  $t$ . For Qwen under the reported protocol,  $|K_0| = 78$ . This is a small held-out set by the standards of NLP evaluation. Larger factual test sets would improve the assessment of sampling variability and provide more stable retention estimates, but expanding the evaluation set alone would not change the composition of the synthetic training stream and would not address generalization to other domains or answer formats. The revised manuscript states: “The evaluation set ( $|K_0| = 78$  for Qwen;  $|K_0| = 46$  for Gemma 3) is small by NLP evaluation standards; results should be interpreted as regime-level tendencies rather than precise quantitative estimates.”

**Answer format and length.** The primary scorer uses bidirectional substring matching to accommodate short-answer variants. Because longer generated answers could spuriously benefit from that permissive criterion, we also report a strict exact-match re-evaluation of all saved predictions.

Absolute retention scores are lower under strict scoring, while the qualitative regime classifications are consistent across both scorers. This sensitivity check is reported in supplementary Table S2. The revised manuscript adds: “Strict exact-match re-scoring of all saved predictions confirms that regime classifications are not an artifact of the permissive substring criterion (supplementary Table S2).”

**Domain and task generalization.** The conclusions are explicitly scoped to short-answer factual recall on TriviaQA. Whether the observed regime structure and dose–response effects generalize to longer answers, multi-hop reasoning, or other domains is an open empirical question that the current experiments do not address. The revised manuscript adds to Section 6: “The evaluation is restricted to short-answer factual questions; results for longer-form generation, multi-hop reasoning, code, or other domains constitute an open empirical question and are outside the scope of this study.”

#### Author action:

1. **Evaluation-set size qualified (Section 3.2).** The sentence “Results should be interpreted as regime-level tendencies rather than precise quantitative estimates” was added to Section 3.2.
2. **Strict re-scoring added (supplementary Table S2).** All saved model predictions were re-scored under strict exact-match. Results are tabulated in supplementary Table S2. The main text states: “Regime classifications are consistent under strict exact-match scoring (Table S2).”
3. **Domain scope bounded (Section 6).** A sentence explicitly limiting the conclusions to TriviaQA short-answer factual questions was added to Section 6.

*Revised manuscript: Section 3.2 (evaluation protocol); supplementary Table S2 (strict scoring); Section 6 (scope and limitations).*

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#### Reviewer #1, Concern #4 — Exposure reduction and model-dependent responses:

*Why does the five-percent exposure reduction stabilize Qwen but not Gemma 3, and does this generalize across ranks and seeds?*

#### Author response:

We sincerely thank the reviewer for this critical observation. This concern directly exposed a substantive methodological problem in the original manuscript that we had failed to identify ourselves: the five-percent exposure pilot (C1/C5) and the pre-registered dose–response experiments (B7, G2) used different evaluation pipelines and are therefore not directly comparable. The reviewer’s question prompted us to withdraw the original C1/C5 inference, to replace it with a properly paired dose–response analysis, and to report the backbone-dependent responses with formal statistical tests and confidence intervals.

**The original comparison was invalid.** C1 and C5 followed different evaluation pipelines, including differences in  $K_0$  construction and data loading. The previously reported 9.0- and 9.4-percentage-point intervention gains cannot therefore serve as evidence of stabilization. Furthermore, the pilot used single-seed evaluations and compared conditions that differed in pipeline construction, not only in the dose level. We do not defend this comparison. The revised manuscript states: “The C1/C5 pilot used a different evaluation pipeline from the pre-registered B7 and G2 experiments; the previously reported gains are withdrawn and replaced by the paired dose–response analysis.”

**The Qwen dose–response evidence (B7).** The revised Qwen evidence comes from Experiment B7: 0%, 10%, 25%, and 50% example removal at  $r = 256$ , with five paired seeds per dose. Relative to no

removal, Generation 10 retention increased by +1.8 pp at 10% removal ( $p = 0.160$ ), +4.9 pp at 25% ( $p = 0.005$ , 95% CI [2.5, 7.2]), and +9.2 pp at 50% ( $p < 0.001$ , 95% CI [6.9, 11.6]). At 50% removal, the Generation 5–10 retention slope was  $-0.15$  pp/generation, compared with  $-1.13$  pp/generation without removal. The attenuation of the later decline is therefore supported for this Qwen configuration and protocol. The revised manuscript states: “At 50% removal (B7, five paired seeds), Generation 10 retention increased by +9.2 pp (95% CI [6.9, 11.6],  $p < 0.001$ ) and the Generation 5–10 slope was attenuated from  $-1.13$  to  $-0.15$  pp/generation.”

**The Gemma 3 dose–response evidence (G2).** The Gemma 3 response was different in kind. In Experiment G2, 50% removal at  $r = 10$  raised Generation 10 retention by +12.9 pp across three paired seeds (95% CI [9.70, 16.10],  $p = 0.001$ ), with the 10% and 25% doses giving +9.8 and +11.4 pp, respectively. The Generation 5–10 slope was  $-0.91$  pp/generation in the 50% arm versus  $-0.91$  at 0%, equal within rounding. The intervention improved endpoint retention without demonstrably slowing later loss. The revised manuscript adds: “On Gemma 3 1B at  $r = 10$ , 50% removal improved Generation 10 retention by +12.9 pp but did not reduce the observed rate of loss ( $-0.91$  versus  $-0.91$  pp/generation).”

**Why the responses differ between backbones.** The reviewer asks whether the dose–response generalizes across ranks and seeds. For Qwen, it does: the B7 experiment uses five paired seeds at  $r = 256$  and shows a graded, statistically significant response. For Gemma 3, the G2 experiment uses three paired seeds at  $r = 10$  (degradative regime) and shows endpoint improvement without rate reduction. Whether the same pattern holds at other ranks or with more seeds is an open empirical question.

#### Author action:

1. **C1/C5 pilot excluded from all estimates.** The 9.0 and 9.4 percentage-point gains previously cited for C1 and C5 are no longer used as evidence of stabilization. The text notes that those runs used a different evaluation pipeline and are excluded from every quantitative estimate in the revision.
2. **Qwen dose–response (B7) reported in Section 4.5.** The B7 results (0%, 10%, 25%, 50% removal, five paired seeds at  $r = 256$ ) are reported in Section 4.5: +1.8 pp at 10% ( $p = 0.160$ ), +4.9 pp at 25% ( $p = 0.005$ ), +9.2 pp at 50% ( $p < 0.001$ ).
3. **Gemma 3 dose–response (G2) reported in Section 4.6.** The G2 results ( $r = 10$ , three paired seeds) are reported in Section 4.6. Endpoint retention improved at 50% removal; the Generation 5–10 slope was equal within rounding, indicating endpoint benefit without demonstrable slope attenuation.
4. **Mechanistic caveat added in Section 4.6.** Section 4.6 notes that the backbone difference may reflect rank or seed-count specifics, and that a causal interpretation across backbones is not supported by the current evidence.

*Revised manuscript: Section 4.5 (Qwen B7 dose–response); Section 4.6 (Gemma 3 G2 dose–response and mechanistic caveat); Section 3.5 (ETP framework and pilot exclusion).*

#### Revised manuscript:

The revised manuscript reports the B7 and G2 results in Sections 4.5 and 4.6 and adds the mechanistic caveat in Section 4.6.

#### Reviewer #1, Concern #5 — Scope of the reproducibility package:

*What complete code, checkpoints, prompts, seeds, and raw outputs will support independent reproduction?*

**Author response:**

We sincerely thank the reviewer for emphasizing the importance of a complete and verifiable reproducibility package. Reproducibility is a foundational requirement for experimental work in machine learning, and the reviewer is correct that aggregate summary tables alone are insufficient to support independent replication. The revision describes precisely what the public package contains, what it does not contain and why, and what a researcher would need to reproduce the headline results.

We agree that aggregate summary tables are insufficient to support independent reproduction, and we accept the reviewer’s implicit requirement that the scope of what is and is not publicly available must be stated precisely and honestly. The revision provides this description in Appendix A.

**What the archive contains.** The Zenodo archive (DOI: <https://doi.org/10.5281/zenodo.23146342>, release v1.0.3) contains: (i) consolidated per-seed results files (one JSON per experimental arm) from which all reported statistics were derived; (ii) the `make_all.py` script that regenerates all tables and numbers in `numbers.tex` from those files; (iii) the figure-generation scripts (including `fig9_dose_response.py`) that produce all reported figures at 300 dpi; (iv) the evaluation protocol, question partition identifiers, and the  $K_0$  baseline sets for both backbones; (v) the training configuration files specifying adapter rank, learning rate, batch size, number of steps, seed, and synthetic data fraction for every reported experimental condition.

**What the archive does not contain, and why.** Raw token-level predictions and trained adapter checkpoints are not included in the archive because they exceed practical size limits for Zenodo deposits and because base-model weights cannot be redistributed under their respective licenses. The archive therefore supports replication of the statistical analysis and figure generation from pre-computed per-seed results, but not a byte-for-byte reconstruction of every training run. The revised Appendix A states explicitly: “[The archive permits reproduction of all reported statistics and figures from the consolidated per-seed results files. It does not include raw predictions, trained adapter checkpoints, or base-model weights, which must be obtained separately.](#)”

**Version discipline.** Scripts introduced or revised after the initial release must be identified by the version in which they are actually deposited. Appendix A now states which scripts are present in release v1.0.3 and which were added for this revision, with the corresponding archive version for each.

**Author action:**

1. **Appendix A rewritten.** The reproducibility appendix was expanded to enumerate what the archive contains (per-seed results, scripts, configurations, evaluation protocol) and what it does not contain (raw predictions, checkpoints, base weights), with the reason for each omission.
2. **Archive contents cited (Appendix A).** The sentence “[The archive permits reproduction of all reported statistics and figures from the consolidated per-seed results files](#)” was added as the opening of Appendix A.
3. **Version discipline documented (Appendix A).** Each script in the archive is now identified by the release version in which it first appears, distinguishing v1.0.3 (initial release) from the current revision.

*Revised manuscript: Appendix A (reproducibility and data availability).*

### Reviewer #2, General assessment — Status of the framework and of the cross-architecture claims:

*The paper addresses a timely and practically relevant question and provides a useful empirical dose-response characterization. However, the current manuscript has major issues in notation, equations, table/figure formatting, internal consistency, statistical rigor, and citation accuracy. The “effective training pressure” framework is promising but currently conceptual rather than quantitative. The cross-architecture claims are weakened by the uncited Gemma 4 E2B model and limited scale. I recommend major revision before further consideration.*

### Author response:

We sincerely thank Reviewer #2 for the detailed and constructive evaluation of the manuscript. The general assessment correctly identifies two substantive problems — overclaiming on cross-architecture generalization and the notation/rendering defects — alongside a range of specific technical issues. We have treated all of these as corrections rather than as matters of perspective, and the revision addresses each point with specific changes to the text, figures, and notation.

We agree with the characterization and have treated all objections as corrections rather than as matters of emphasis. The two substantive issues — the conceptual rather than quantitative status of the ETP framework and the weakness of the cross-architecture claims — are addressed first. The specific technical corrections are addressed in the individual responses below.

**On the status of the ETP framework.** The reviewer is correct that effective training pressure, as submitted, was a conceptual label for three already-known axes and not a validated quantitative measure. We do not defend the original framing. The revised manuscript corrects this at every level. No scalar predictor is derived from the framework. The contribution is restated as: (i) a factorial experimental design that simultaneously varies adapter rank, learning rate, and synthetic data dose while holding other factors constant; (ii) pre-registered dose-response evidence on two backbones showing that the regime boundary can be shifted by reducing synthetic exposure; and (iii) the empirical finding that the regime boundary location is backbone-dependent and is not reconciled by any architectural normalization we tested. We do not claim to have derived a transferable quantitative predictor. The text now states explicitly: “[The effective training pressure framework organizes the three independently manipulated axes in this study. It is not a new quantitative measure and does not produce a transferable predictor; the cross-architecture normalization did not align the observed boundaries, and we report this failure rather than minimizing it.](#)”

**On the cross-architecture claims and Gemma 4 E2B.** The reviewer correctly identifies that Gemma 4 E2B IT lacks a citation and may be confused with Gemma 3n E2B. This has been corrected. Reference [41] (Gemma Team, arXiv:2607.02770, 2026) is now included for Gemma 4 E2B IT, and the model is explicitly distinguished from Gemma 3n E2B in both the text and the conditions table (Table 2). Because Gemma 4 E2B IT was evaluated with a single seed, its results are labeled supportive validation and excluded from every threshold estimate and from all multi-seed statistical analyses. The revised manuscript states: “[Gemma 4 E2B IT \(single seed; supportive validation only\) is evaluated for regime-level consistency but excluded from threshold estimates and multi-seed statistical comparisons; see Reference \[41\] for the model description.](#)”

**On the scale limitation.** The restriction to 1–2 B parameter models is acknowledged as the first limitation in Section 6: “[All experiments use 1–2 B parameter models; the regime structure and](#)

pressure thresholds at larger scales (7B+) are not established and constitute the most important direction for future work.”

**On the “confirms” language.** The reviewer’s objection to the word “confirms” in one place prompted us to conduct a systematic audit of epistemic language throughout the manuscript. All uses of “confirms,” “demonstrates,” and “proves” were reviewed; those not supported by multi-seed confirmatory evidence were replaced with “is consistent with,” “suggests,” or “indicates.” This audit is described in detail in the response to Reviewer #2, Concern #4.

Beyond the points listed below, the reviewer’s objection to the word “confirms” in one place prompted us to apply the same standard everywhere. We re-read the Results and Discussion for claims whose strength exceeded the design supporting them, and corrected fifteen further instances: a single-seed grid described as *demonstrating* joint dependence, now reported as descriptively consistent with it; a degradative regime described as *reversible*, now described as attenuated to the bounded regime on one backbone and as a retention gain without slope reduction on the other; a 50 % exposure reduction described as *arresting* progressive loss, now quantified as reducing its rate to  $-0.15$  pp/generation; a single-seed observation on a third backbone described as *generalising*, now reported as consistent with but not establishing generalization; and a non-monotonic slope response across doses that the original text had smoothed, now reported as two distinct outcomes. We also found one paragraph in Section 5.4 that still drew a conclusion from the archived pilot described in our response to Reviewer 1, Concern #4, and removed it.

*Revised manuscript: §1; §3.3; §3.5; §4.1; §4.3.2; §5.1–5.4; §6.*

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## Reviewer #2, Concern #1 — Abbreviations and notation:

*ETP appears in Figure 1 but is never explicitly defined as “effective training pressure.” MTLD is introduced without expansion. KL, JS, and coverage appear in Figure 1 without definition. bf16, SVD, TCE, QLoRA, and LoRA are used without consistently expanded forms. “Homeo.” is used in Table 2 without definition.  $K_0$  is used both as a set of questions and as a retention metric.  $T$ ,  $D1\_T$ ,  $D10$ ,  $\Delta I$ ,  $Instability(1)$ ,  $MeanLen\_T$ , and related symbols in Eq. 6 are not fully defined. The effective-rank formula uses  $\sigma_i$  as both singular values and normalized singular values. The normalization condition is mathematically incorrect as written.  $BA\_F$ ,  $\theta_i^{(t)}$ ,  $\theta_i^{(0)}$ , and related notation in the drift equation are not consistently defined.  $r \in \{4, 10, 12, 14, 16\}$  set notation was garbled in the original PDF.*

### Author response:

We sincerely thank the reviewer for this observation. Self-contained notation is an essential condition for a manuscript that introduces several non-standard abbreviations (ETP, SDI-3, the regime labels) and a complex factorial design. The reviewer correctly identified that ETP appeared in Figure 1 before it was defined in the text, and that several notational choices — particularly the disambiguation between generation subscripts and  $n$ -gram orders in the diversity metrics — were insufficiently clear. The revision addresses each point directly.

We agree that the original notation was insufficiently self-contained, and we acknowledge that this is a significant usability defect for a manuscript that introduces several non-standard abbreviations alongside a complex factorial design. We address each notational issue raised by the reviewer in turn.

**ETP and Figure 1.** The abbreviation ETP appeared in Figure 1 before it was defined in the text. This has been corrected: Figure 1 no longer carries ETP unexpanded, and the term is introduced

in full (“effective training pressure”) at its first use in Section 1. The abbreviation is then formally defined in Section 3.5, where it is needed for the compound labels ETP-design and ETP-observed. The revised Figure 1 caption states: “Stage 3: Pressure manipulation along five axes (nominal rank, learning rate, method, module topology, and synthetic-data dose); see Section 3.5 for the effective training pressure (ETP) framework.”

**Unexpanded abbreviations.** All abbreviations listed by the reviewer have been expanded at first use: Measure of Textual Lexical Diversity (MTLD) in Section 3.6; bfloat16 (bf16) in Section 3.1; singular value decomposition (SVD) in Section 3.5.1; Truncated Cross-Entropy (TCE) in Section 3.1; Low-Rank Adaptation (LoRA) in Section 1; and its quantized variant QLoRA in Section 1. “Homeo.” in Table 2 has been replaced with “Homeostatic” throughout.

**KL, JS, and coverage in Figure 1.** These three metrics appeared in the original Figure 1 but are not computed in the reported experiments. They have been removed from Figure 1 entirely and replaced with the metrics that are actually computed: Distinct- $n$ , MTLD, and stopword ratio. The revised Figure 1 caption now lists: “Stage 4: Per-generation evaluation of factual retention ( $K_0$ ), effective rank, response length, Distinct- $n$ , MTLD, and stopword ratio.”

**$K_0$  dual use.** The symbol  $K_0$  was used both as a set (the questions answered correctly at generation 0) and informally as a retention metric. The revision separates these:  $K_0$  now denotes only the set,  $|K_0|$  its cardinality, and  $R(t)$  the retention ratio at generation  $t$  as defined in Eq. 1.

**SDI-3 notation ( $T$ ,  $D1\_T$ ,  $D10$ , etc.).** The notation in Eq. 6 has been rewritten.  $T$  now denotes the index of the final evaluated generation (Generation  $T$ ),  $D1_t$  the Distinct-1 diversity metric at generation  $t$ , and  $D1_0$  the baseline value at generation 0. The subscript convention now uses the generation index in subscript position to disambiguate from the  $n$ -gram order, removing the ambiguity between  $D10$  (generation index 10) and  $D_{10}$  (Distinct-10). The  $\Delta I$  term is defined inline with Eq. 6 as the difference in mean response length between generation  $T$  and generation 0.

**Effective-rank formula ( $p_i$  notation).** The original Eq. 2 used  $\sigma_i$  both for the raw singular values and for the normalized weights. The revision uses  $p_i$  for the normalized weights, with  $p_i = \sigma_i / \sum_j \sigma_j$ , making the normalization condition  $\sum_i p_i = 1$  explicit and unambiguous.

**Drift equation ( $BA_F$ ,  $\theta_i^{(t)}$ ,  $\theta_i^{(0)}$ ).** All symbols in the drift equation are now defined in the text immediately preceding the equation:  $BA$  denotes the composed adapter update matrix,  $\|\cdot\|_F$  the Frobenius norm, and  $\theta_i^{(t)}$  the  $i$ -th parameter of the model at generation  $t$  with  $\theta_i^{(0)}$  denoting the frozen base-model value.

$r \in \{4, 10, 12, 14, 16\}$  **garbling.** This was a rendering artefact of the original PDF (see the response to Concern #2 below). The set notation is now correct in the revised PDF.

#### Author action:

We revised Figure 1, the notation table, first-use acronym definitions, the regime-table labels, and the text preceding Eq. 6.

*Revised manuscript: Figure 1 and caption; notation and regime tables; Section 3.5; definitions surrounding Eqs. 1 and 6.*

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#### Reviewer #2, Concern #2 — Equations and tables:

*Equations 1-6 are severely malformed in the provided PDF. For example, Eq. 1 uses “Retention.t /” instead of a clear function notation and has garbled summation over  $K_0$ . Eq. 2 for effective rank is missing the negative sign and summation. It should be*

something like  $\exp(-\sum p_i \log p_i)$ , with  $p_i = \sigma_i / \sum_j \sigma_j$ . Eq. 6 for SDI-3 is unclear: signs, log ratios, and the  $\Delta I$  term are not consistently defined. What is  $5 \times 10^6$  in Table 7? Please check also elsewhere.

#### Author response:

We sincerely thank the reviewer for this detailed audit of the equations and tables. The rendering defects identified by the reviewer (garbled summation in Eq. 1, missing negative sign in Eq. 2, unclear terms in Eq. 6) are serious, and we take full responsibility for not having detected and resolved them in the original submission. We have traced the root cause to the re-distillation of embedded font subsets by the Editorial Manager, corrected all mathematical errors identified, and changed the font pipeline to prevent recurrence of the rendering fault.

Two of these points are defects in the mathematics and two are defects in the rendering, and we treat them separately because they have different causes.

*The rendering fault.* Rather than re-render the document and hope for a different outcome, we traced it. Two audits establish where it does *not* originate. First, an automated scan of the complete L<sup>A</sup>T<sub>E</sub>X source returns a single non-ASCII character, inside a comment: no mathematical symbol is encoded as raw Unicode, and the source is syntactically correct as submitted. Second, `pdffonts` confirms that every font in our locally compiled PDF is embedded with a complete `ToUnicode` mapping, and text extraction from that file reproduces the equations correctly. The corruption is introduced when the uploaded file is re-processed: the embedded fonts use a custom encoding, and the re-distillation step does not preserve their Unicode mappings, so that minus signs, multiplication signs, set-membership symbols and percent signs are drawn correctly on the page but map to unrelated code points on extraction. This accounts for the specific form of the corruption the reviewer observed (“Retention.t /”, “5 × 10<sup>6</sup>”, “r E 4,10,12,14,16”) rather than missing or blank glyphs. For this revision the font pipeline no longer relies on custom-encoded subsets. Following the reviewer’s instruction to check elsewhere, we also wrote an audit that compares the extracted text of every equation, table cell, figure label and caption against the L<sup>A</sup>T<sub>E</sub>X source; its script and output are included in the replication package.

*The mathematical defects.* Equation 2 was wrong as written, not merely mis-rendered, and we have adopted the reviewer’s formulation. The original text used  $\sigma_i$  for both the singular values and their normalization, which makes the normalization condition circular. The revision introduces a separate symbol for the normalized weights and states the condition explicitly. Equation 6 was underspecified: the sign convention and the direction of the diversity ratio were not stated, so the sign of SDI-3 could not be interpreted. All three terms are now defined, and the direction of the composite is stated.

| Point raised                     | Correction                                                                                                                                                                                                                                                                          |
|----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) Eq. 1 notation and summation | <p>Rewritten with explicit function notation and set cardinality:</p> $R(t) = \frac{ \{q \in K_0 : q \text{ answered correctly at generation } t\} }{ K_0 }$ <p>The numerator is the cardinality of a set, not a summation, which is what the garbled original obscured. (§3.2)</p> |

- (b) Eq. 2 missing sign and summation; normalization incorrect

Accepted as a mathematical error and corrected in the notation the reviewer proposed:

$$\text{erank}(BA) = \exp\left(-\sum_i p_i \log p_i\right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j}$$

with “ $p_i \geq 0$  and  $\sum_i p_i = 1$ ” stated in Table 1. The symbol  $p_i$  denotes the normalized singular weight;  $\sigma_i$  are the raw singular values of  $\Delta W$ . (Eq. 2, §3.5.1; Table 1)

- (c) Eq. 6 signs, log ratios,  $\Delta I$

Rewritten with each ratio oriented so that all three terms increase under deterioration:

$$\text{SDI-3} = \log \frac{\text{MeanLen}_T}{\text{MeanLen}_0} + \log \frac{\text{D1}_0}{\text{D1}_T} + \Delta I$$

with  $\Delta I = \text{Instability}(T) - \text{Instability}(1)$ , the convention  $0 \log 0 = 0$ , and the interpretive statement “a positive SDI-3 score indicates cumulative distributional deterioration”. Each term is named in the preceding prose: verbosity increase, unigram diversity decrease, loss of item-level response stability. (Eq. 6, §3.6)

- (d)  $5 \times 10^{-6}$ ; check elsewhere

The value is the learning rate  $5 \times 10^{-6}$  in Table 7 of the original submission (now Table 9); both the multiplication sign and the negative exponent were lost to the glyph fault. The instruction to check elsewhere was followed systematically: the audit covers every equation, table cell, figure label and caption, and also recovered the garbled set expressions, now rendered as  $r \in \{4, 10, 12, 14, 16\}$  and  $r \in \{10, 12, 14\}$ . (Table 9; §3.4)

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#### Author action:

1. **Font pipeline changed.** No embedded subset uses a non-standard encoding. The revised PDF uses `cmap` and `glyphtounicode` to maximize Unicode coverage; a residual number of Latin Modern Math Type 1 operators lack ToUnicode entries (a known limitation of that font subset) and are flagged in the audit output, which is included in the replication package. The fault was verified at the point where it appears, not only locally.
2. **Equations 1, 2, 3 and 6 rewritten (Section 3.2 and 3.5–3.6).** Each was cross-checked against Table 1 term by term. Eq. 1: “ $R(t) = |K_0 \cap A(t)|/|K_0|$ ”. Eq. 2: “ $\text{erank}(M) = \exp(-\sum_i p_i \log p_i)$ ,  $p_i = \sigma_i / \sum_j \sigma_j$ ”. Eq. 3 now shows absolute value and summation explicitly. Eq. 6 rewritten with all terms defined inline.
3. **Glyph audit script included in replication package.** `audit/glyph_audit.py` included in the package so the check can be repeated independently.
4. **Every figure regenerated at 300 dpi.** Numerical values plotted in each figure were re-verified against the corresponding table; no figure reports a value different from its table.

*Revised manuscript: §3.2; §3.4; §3.5.1; §3.6; Table 1; Table 9; replication package (audit script and output).*

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## Reviewer #2, Concern #3 — Technical and data inconsistencies:

(a) *Qwen threshold inconsistency:* Abstract and Section 4.2.3 say the boundary is between effective ranks 50–88; Section 5.1 says Qwen remains outside the degradative regime “up to effective rank 150.” This contradicts Table 2, where  $r = 256$  has effective rank 87.57 and is degradative. (b)  $r = 128$  is classified as “Bounded” in Table 2, but Section 4.4.3 calls  $r = 128$  and  $r = 256$  “above-threshold configurations.” (c)  $r = 256$  is repeatedly called a “boundary configuration”, but Table 2 classifies it as degradative. (d) *Gemma 3 degradative range:* Section 5.2 says “ $r \geq 16$  on Gemma 3,” but Table 4 shows  $r = 10, 12, 14$  already degradative at Gen 5. (e) *Abstract overclaim:* “three to five independent seeds per condition” is not true. (f) *FFT on Gemma 3 shows non-monotonic results;* the conclusion that this “confirms” the magnitude axis is too strong. (g) *C5 removal percentage:* Section 3.4 mentions “15% removal,” while Section 4.5 reports approximately 5% token reduction.

### Author response:

We sincerely thank the reviewer for the systematic and rigorous audit of the technical claims and numerical consistency of the manuscript. All seven inconsistencies identified are genuine errors or overclaims: each is a statement in the manuscript that did not match the data as reported in the tables, or that made a claim stronger than the evidence supports. We accept every point without reservation and have corrected each one in the revised manuscript. We are grateful for the level of attention to detail that this review demonstrates.

All seven are accepted. They are not matters of emphasis: each is a statement in the manuscript that did not match the design or the table supporting it. We answer them individually and, where the correction changes a claim rather than a word, say so.

(a) *Qwen threshold stated inconsistently.*

The phrase “up to effective rank 150” was wrong and has been removed. The reviewer’s reading of Table 4 is correct:  $r = 256$  has mean effective rank 87.57 and is degradative, so no claim of homeostasis beyond that value can stand. The deeper problem the reviewer exposes is that we were using a single figure to describe two different boundaries. The revision distinguishes them. On Qwen, the tested grid places the transition from homeostatic to bounded behavior between effective ranks 29.5 ( $r = 64$ ) and 50.2 ( $r = 128$ ), and the transition from bounded to degradative between 50.2 and 87.6 ( $r = 256$ ). The “50–88” interval quoted in the Abstract is the second of these. On Gemma 3, where no bounded regime was observed, the “3–6” interval is the transition from homeostatic directly to degradative. The two intervals are therefore not the same kind of object, and the revision no longer compares them as though they were. “(§4.2.3; §5.1; Abstract)”

(b)  $r = 128$  labeled above-threshold while classified Bounded.

Corrected. “Above-threshold configurations” now refers exclusively to  $r = 256$ . The  $r = 128$  configuration is described as a “transition-zone configuration”, in which distributional signatures emerge without a progressive retention trajectory, which is precisely the dissociation we report for it, and which the original label obscured. “(§4.4.3; §5.3)”

(c)  $r = 256$  called a boundary configuration.

Corrected throughout. The reviewer is right that the boundary lies between  $r = 128$  and  $r = 256$ , not at  $r = 256$ . Every occurrence now reads “lowest-pressure degradative configuration tested”, which states what the configuration is without implying that it sits on the boundary. “(§3.4; §4.5; §5.4)”

(d) *Gemma 3 degradative range.*

Corrected. Table 4 of the original submission (now Table 6) reports 78.3%, 73.9% and 69.6% at  $r = 10, 12$  and  $14$  respectively, all degradative at Gen 5, so “ $r \geq 16$ ” was inconsistent with our own data. The text now reads “ $r \geq 10$  on Gemma 3” and locates the boundary “below  $r = 10$ ”, between effective ranks 3 and 6. “(§5.2; Table 6)”

(e) *Seed count in the Abstract.*

Corrected, and the correction goes beyond the Abstract. The seed policy is now stated at the level of individual claims rather than once in aggregate: headline conditions use three or five seeds (six for the pooled  $r = 256$  estimate), intermediate ablations use a single seed and are labeled descriptive, and the two classes are never pooled when a headline claim is made. An  $N$  column was added to the rank-sweep and dose-response tables; the remaining tables (Tables 7–9) state  $N$  in a footnote. Contribution 1 in the Introduction, which carried the same overclaim, was corrected with it. “(Abstract; §1; §3.8; Tables 4–10)”

(f) *FFT on Gemma 3 described as confirmatory.*

Accepted. A single-seed, non-monotonic result cannot confirm anything, and “confirms” has been replaced by “is consistent with”. The non-monotonicity is now reported rather than smoothed over: Gen 5 retention was 78.3% at  $\eta = 10^{-6}$ , 54.3% at  $5 \times 10^{-6}$ , and 63.0% at  $10^{-5}$ , and the text states that “the non-monotonic pattern at  $LR = 5 \times 10^{-6}$  limits interpretation. We treat this as descriptive evidence rather than confirmatory generalization.” Contribution 2 in the Introduction carried the same overclaim and was corrected. “(§1; §4.3.3)”

(g) *C5 removal percentage.*

The reviewer identified a genuine ambiguity, and the revision removes it rather than reconciling it. Two things were conflated: the fraction of training *examples* removed and the resulting reduction in *tokens*, which differ because responses vary in length. More substantively, the comparison those figures supported is no longer used: the C1–C5 pilot was found to compare arms produced by different evaluation pipelines and is excluded from every estimate in the paper (see Reviewer 1, Concern #4). In the two pre-registered experiments that replace it, exposure is specified exclusively as the fraction of training examples removed — 0%, 10%, 25%, 50% — with no token-based figure anywhere. The pilot, including its original dose description, is archived in the replication package. “(§3.7; §4.5; Figure 9)”

**Author action:**

1. **Qwen threshold (a) corrected (Section 5.1).** “Up to effective rank 150” deleted. Two distinct boundaries stated: “homeostatic-to-bounded: effective ranks 29.5–50.2; bounded-to-degradative: 50.2–87.6.” Corrected systematically, not selectively.
2. **Regime labels (b, c) corrected.**  $r = 128$ : “transition-zone configuration ( $r = 128$ )”;  $r = 256$ : “lowest-pressure degradative configuration tested ( $r = 256$ )” at every occurrence.
3. **Gemma 3 range (d) corrected (Section 5.2).** “ $r \geq 10$  on Gemma 3” (was “ $r \geq 16$ ”), consistent with Table 4.
4. **Seed count (e) corrected and  $N$  column added.** Abstract: “Most headline conditions use three to five seeds.”  $N$  column added to rank-sweep and dose-response tables; remaining tables state  $N$  in a footnote.
5. **“Confirms” (f) replaced; non-monotonic values reported.** “is consistent with” in Contribution 2 and §4.3.3; non-monotonic values stated explicitly.

6. **Token/example ambiguity (g) removed.** All dose specifications restricted to fractions of training examples; no token percentage used.
7. **Three further instances corrected.** A systematic re-reading found three additional instances where a horizon (Gen 5 vs. Gen 10) or seed count was implied but not stated; all corrected.

*Revised manuscript: Abstract; §1; §3.4; §3.7; §3.8; §4.2.3; §4.3.3; §4.4.3; §4.5; §5.1–5.4; Tables 4–10.*

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## Reviewer #2, Concern #4 — Methodology:

*No formal statistical tests, confidence intervals, or p-values are reported. The effective training pressure framework is not quantified. Cross-architecture normalization failed, but no alternative calibration method is proposed. “Effective training pressure” is not a new quantitative measure; it is a conceptual label for three already-known axes. The Gemma 4 E2B backbone is a serious concern: it lacks a citation, may be confused with Gemma 3n E2B, and its architecture is described without a source. The “sharp transition” claim is weakened by sparse rank grids and wide threshold intervals.*

### Author response:

We sincerely thank the reviewer for this thorough methodological critique. The absence of formal statistical tests and confidence intervals is a significant limitation of the original manuscript, and the reviewer is correct that aggregate patterns alone are insufficient to distinguish true effects from noise in small-sample experimental designs. The other sub-concerns — the use of pre-registered versus exploratory analyses, the confounding of multiple experimental changes, and the language of overclaiming — are equally important. We address each sub-concern in turn and have made corresponding changes to the manuscript.

#### (a) *No formal tests, confidence intervals, or p-values.*

Inferential statistics have been added wherever the design admits them, and the design is now stated alongside each test rather than left to be inferred. For the Gemma 3 anchor-rank contrast ( $N = 5$  per rank,  $|K_0| = 46$ ), the mean difference is +37.8 pp; because the archived data do not confirm whether seeds were paired across ranks, we report both possibilities — an exact sign-flip test at  $p = 0.0625$  if paired, a bilateral permutation test over  $\binom{10}{5} = 252$  allocations at  $p = 0.0079$  if independent — with Hedges’  $g \approx 13.3$  under either reading. For the Qwen rank contrast, the clean comparison is the G3 paired design (same seeds, same precision, rank the only variable): per-seed differences of +15, +13 and +15 items out of 78, mean +18.4 pp, exact sign test  $p = 0.25$  at  $N = 3$ . We report that  $p$ -value rather than suppress it: the direction is consistent across all three pairs, but three pairs cannot produce a small  $p$ , and we say so. For the two dose–response experiments, paired  $t$ -tests with 95 % confidence intervals are reported at every dose level. Where groups are not exchangeable under the relevant null — the 17.7 pp gap between the three-seed  $r = 16$  condition and the six-seed  $r = 256$  estimate, which differ in rank, precision protocol and seed assignment simultaneously — the comparison is labeled descriptive and no test is reported. “(§4.1; §4.2.1; §4.5; §4.6)”

#### (b) *The framework is not quantified.*

Correct, and the revision no longer implies otherwise. No scalar predictor is derived. The manuscript now separates the variables fixed before training — nominal rank, learning rate, training steps, dose fraction — from the quantities measured after it — effective rank, weight drift, response

length, SDI-3 — and states that the latter “are not available prospectively; they therefore serve as post-hoc explanatory variables rather than pre-specified predictors”. The operational index  $\Pi = \text{erank}(\Delta W) \cdot (\eta/10^{-5})$  is labeled “an exploratory post-training summary” and is “not used as a validated quantitative threshold”. Developing a quantitative formulation is listed as future work, with the protocol a prospective test would require specified in Section 6. “(§3.5; §6)”

(c) *Normalisation failed with no alternative proposed.*

The reviewer asks for the constructive half of a negative result, and the original manuscript supplied only the negative half. Five normalizations were tested — by hidden dimension, its square root, layer count, attention heads, and key–value heads — and none aligned the thresholds across backbones (Table 13). We retain that table as a reported negative result and now propose what we believe the finding implies: that a transferable constant should not be sought, and that calibration must be per-backbone. Section 6 specifies the procedure concretely — a pilot sweep of two ranks spanning roughly an order of magnitude, five generations, single seed; the boundary estimated as the interval between the highest configuration whose early retention slope is indistinguishable from zero and the lowest whose slope is negative — together with its cost, which is approximately two short runs per backbone. We present this as the practical alternative to a universal normalization rather than as a validated method, since we have not executed it on an unseen backbone. “(§4.2.3; §6; Table 13)”

(d) *Not a new quantitative measure.*

We agree, and we do not defend the original framing. Effective training pressure, as submitted, was a conceptual label for three axes already known individually, and presenting it as a measure was an overstatement. The revision restates the contribution as what the experiments actually establish: a factorial design over three axes; dose–response evidence along each; a retention/distribution dissociation at intermediate capacity that is not captured by existing accounts of recursive collapse; two pre-registered exposure experiments, one per backbone; and the finding that the regime boundary is backbone-dependent by roughly an order of magnitude and is not reconciled by any architectural normalization we tested. The organizing value of the framework is that three independently manipulated variables move the system across the same qualitative boundary, which is an empirical observation and is presented as one. “(§1; §3.5; §5.1; Table 11)”

(e) *Gemma 4 E2B: citation, possible confusion with Gemma 3n E2B, unsourced architecture.*

All three parts are addressed. The backbone is now cited as Gemma Team [41] at every point of introduction and description. On the possible confusion, which we should have pre-empted: the model is Gemma 4 E2B IT, a distinct release from Gemma 3n E2B, and the revision states this explicitly rather than relying on the reader to infer it from the version number. The architectural description is attributed to the released model configuration rather than asserted — 35 layers, hidden size 1536, grouped-query attention with 8 attention heads and 1 key–value head, and key–value projections shared across the final 20 of 35 layers, which is why a LoRA adapter targeting `q_proj` and `v_proj` adapts 35 query and 15 value matrices. The repository identifier and revision hash are listed in supplementary Table S1. We also constrain what the backbone is used for: it was evaluated with a single seed, so it is labeled supportive validation, excluded from every threshold estimate, and the limitation is stated in those terms — “they do not constitute an independent replication of the quantitative threshold estimates”. “(§3.3; §4.2.2; §6; Table S1)”

(f) *“Sharp transition” weakened by sparse grids and wide intervals.*

Accepted, and the correction is larger than a word substitution. “Sharp” has been removed, but so has the argument that supported it. The original Section 5.1 offered the contrast between a 10 % and a 50 % exposure reduction as evidence of abruptness; those doses are not adjacent, so the contrast cannot speak to the steepness of a boundary, and the argument has been withdrawn. What

the data support is weaker and is now stated as such: within the tested grid, retention changes between adjacent configurations in a manner consistent with a threshold, and the location of that change can be bracketed but not resolved. The revision also quotes the grid spacing wherever a boundary is described, so that the reader can judge the resolution: on Qwen, adjacent tested ranks double ( $r = 64, 128, 256$ , giving effective ranks 29.5, 50.2, 87.6), and on Gemma 3 the relevant interval spans effective ranks 3.07 to 6.37. The phrase used throughout is “a threshold-like transition within the tested grid”, and Section 5.1 now states that “the present experiments do not establish a precise threshold value that will generalize to other settings”. “(Abstract; §1; §5.1; §7)”

#### Author action:

1. **Formal tests added (a) (Sections 4.1–4.5).** Exact permutation and sign-flip tests for rank contrasts; paired  $t$ -tests with 95 % CIs for all dose levels; Hedges’  $g$  for the Gemma 3 anchor contrast. Each reported with  $N$  and pairing assumption stated. “We report exact  $p$ -values regardless of conventional thresholds; small  $N$  limits power even when direction is fully consistent across all seeds.”
2. **ETP reframed (b) (Section 3.5).** Restructured around the design/observed distinction. “ETP is not a new quantitative measure; it is an organizational label for three independently manipulated axes.”
3. **Calibration procedure added (b) (Section 6).** Five-step per-backbone calibration proposed as the alternative to universal normalization, with cost stated.
4. **Contribution statement rewritten (Section 1 and 5.1).** Claims the factorial design and the evidence, not a measure.
5. **Gemma 4 cited (e) (Section 3.3 and Table 2).** Cited as Ref. [41], distinguished from Gemma 3n E2B, labeled supportive validation (single seed; excluded from multi-seed analyses).
6. **“Sharp”/“abrupt” replaced (f) (throughout).** All nine occurrences of *sharp* have been replaced with “threshold-like transition within the tested grid”; non-adjacent-dose argument withdrawn; grid spacing quoted.

*Revised manuscript: §1; §3.3; §3.5; §4.1; §4.2.1–4.2.3; §4.5–4.6; §5.1; §6; §7; Table 11; Table 13; Table S1.*

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#### Reviewer #2, Concern #5 — Other format problems:

*Reference [12] and [33] share the same arXiv number: arXiv:2408.14572. This is a serious citation error. Reference [32] lists arXiv:2512.15634 with year 2024, which is impossible if the arXiv ID is December 2025. No citation is provided for Gemma 4 E2B IT, despite it being a central backbone. Reference [7] “Keisha et al.” is used heavily but appears to be a preprint with no venue. Its reliability should be clarified. Reference formatting is inconsistent: some entries include conference proceedings, others only arXiv, some lack page numbers or venues.*

#### Author response:

We sincerely thank the reviewer for the careful verification of the bibliography, reference formatting, and source attributions. The duplicate arXiv identifier, the missing identifiers for preprints, and the label assigned to reference [41] are all genuine errors that should have been caught before the original submission. We accept all five points and have corrected each in the revised bibliography.

All five are accepted and corrected. The first three were errors of fact in the bibliography; the fourth concerns how a source is characterized in the text; the fifth required an audit of the whole list.

| Point raised                                     | Correction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a) [12] and [33] share arXiv:2408.14572         | Two distinct works had been conflated. The identifier belongs to [33] Fawi, <i>CURLoRA: Stable LLM Continual Fine-Tuning and Catastrophic Forgetting Mitigation</i> , arXiv:2408.14572 (2024), which is cited in §2.4 for structural matrix constraints. [12] is Shabgahi et al., <i>ForTIFAI: Fending off Recursive Training Induced Failure for AI Model Collapse</i> , arXiv:2509.08972 (2025), cited in §2.3 for Truncated Cross-Entropy. The identifier now appears exactly once in the bibliography.                                                                                                                                                                                                                                        |
| (b) [32] year 2024 with a December 2025 arXiv ID | Corrected to 2025. The entry is Rathore et al., <i>How Much Is Too Much? Exploring LoRA Rank Trade-offs for Retaining Knowledge and Domain Robustness</i> , and the venue — AACL-IJCNLP 2025 — is now given alongside the preprint identifier, which resolves the inconsistency the reviewer detected rather than merely changing the digit.                                                                                                                                                                                                                                                                                                                                                                                                      |
| (c) No citation for Gemma 4 E2B IT               | Added as [41] Gemma Team, <i>Gemma 4 Technical Report</i> , arXiv:2607.02770 (2026), cited at every point where the backbone is introduced or described. The model is also explicitly distinguished from Gemma 3n E2B in §3.3, and its architectural description is attributed to the released configuration, with the repository identifier and revision hash in supplementary Table S1 (see Concern #4(e)).                                                                                                                                                                                                                                                                                                                                     |
| (d) [7] is a preprint with no venue              | The reviewer is right to ask, and we have made the status visible rather than implicit. The bibliography entry now carries “Non-peer-reviewed preprint” explicitly, and §2.2 marks the comparison as “suggestive context rather than a controlled comparison”, noting that the different dataset, evaluation format and training protocol preclude direct quantitative comparison. We also reduced the load the source carries: no quantitative claim in the paper rests on it, and Shumailov et al. [6] ( <i>Nature</i> ) is the peer-reviewed anchor for the model-collapse phenomenon. [7] is retained because it is the closest prior work on factual degradation under recursive training and omitting it would misrepresent the literature. |

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|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (e) Inconsistent reference formatting | The full list was audited and harmonized to one convention: works published at a venue carry the venue, year and, where applicable, volume and pages; preprints carry the arXiv identifier and year; works that are both carry the venue with the identifier appended, as for [32]. Entries that were previously incomplete were completed against the published records. Three entries not mentioned by the reviewer were also corrected: [31] Soutif-Cormerais et al. (not Biderman et al.), <i>An Empirical Analysis of Forgetting in Pre-trained Models with Incremental Low-Rank Updates</i> , arXiv:2405.18069 (2024); [20] Xu et al., <i>A Probabilistic Perspective on Model Collapse</i> , arXiv:2410.12952 (2025), was missing its arXiv identifier inline; and one further entry had an incomplete venue field. <code>cas-refs.bib</code> is included in the submission package. |
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#### Author action:

1. **Duplicate identifier corrected (a).** arXiv:2408.14572 → Ref. [33]=Fawi (CURLoRA, 2024); Ref. [12]=Shabgahi et al. (ForTIFAI, arXiv:2509.08972, 2025). All identifiers checked programmatically for uniqueness.
2. **Reference [32] corrected (b).** Year 2024→2025; venue AACL-IJCNLP 2025 added: “[Rathore et al., arXiv:2512.15634, AACL-IJCNLP 2025.](#)”
3. **Gemma 4 E2B IT cited (c).** Added as Ref. [41]: “[Gemma Team, “Gemma 4 Technical Report,” arXiv:2607.02770, 2026.](#)”
4. **Reference [7] qualified (d).** Preprint status declared; “(preprint; not peer-reviewed at time of writing)” added in text.
5. **Bibliography audited (e).** Every entry checked against published record for venue, year, volume, pages, and identifier. Three additional corrections found and applied: Refs [20] and [31] gained arXiv IDs; Ref. [31] attribution corrected to Soutif-Cormerais et al., arXiv:2405.18069.

*Revised manuscript: cas-refs.bib (full list); entries [6], [7], [12], [20], [31], [32], [33], [41]; §2.2; §2.3–2.4; §3.3; Table S1.*